SC Honors Geometry · Nappier
5.1 Angles of Triangles · 5.2 Congruent Polygons · 5.3 Triangle Rigidity

NAME _____ DATE _____ PERIOD _____

5.1–5.3 Practice

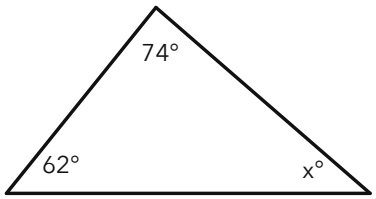
37 figures · quiz Friday · figures are not drawn to scale

PART 3 ONWARD, WRITE ONE OF **SSS** **SAS** **ASA** **AAS** **HL** **NOT ENOUGH**

Part 1. 5.1 — Find the angle

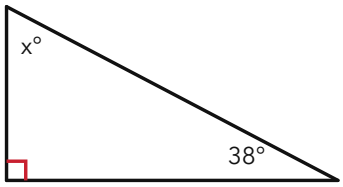
Use the Triangle Sum Theorem or the Exterior Angle Theorem. Solve for *x*.

1.



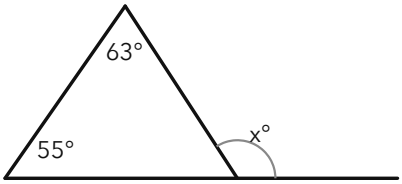
x = _____

2.



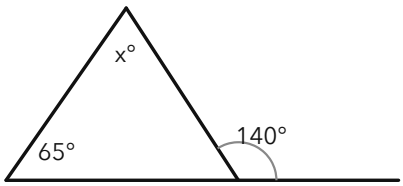
x = _____

3.



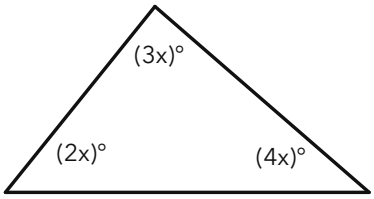
x = _____

4.



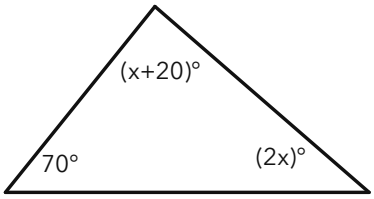
x = _____

5.



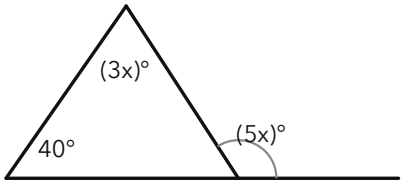
x = _____

6.



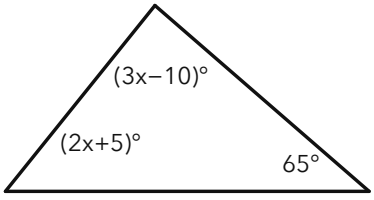
x = _____

7.



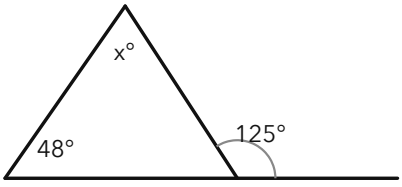
x = _____

8.



x = _____

9.

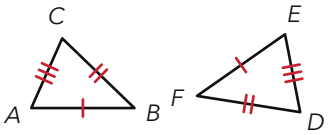


x = _____

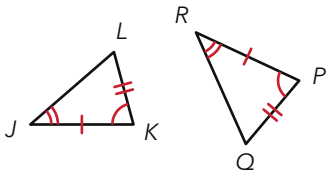
Part 2. 5.2 — Corresponding parts

The marks tell you which vertex goes with which. Answer in matching order.

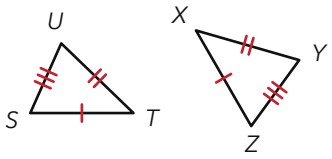
10.



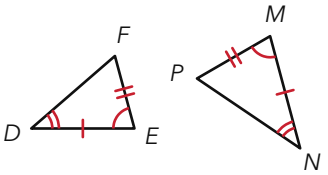
11.



12.



13.

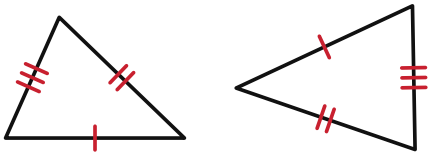


△ABC ≅ △ _____ △JKL ≅ △ _____ Side that matches ST: _____ Angle that matches ∠F: _____

Part 3. 5.3 — Name the criterion

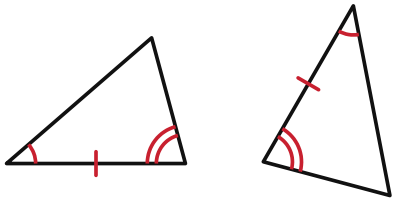
Each pair is marked. Write the criterion that proves the two triangles congruent – or write NOT ENOUGH.

14.



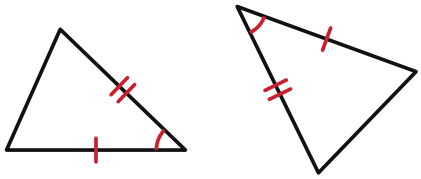
Criterion: _____

15.



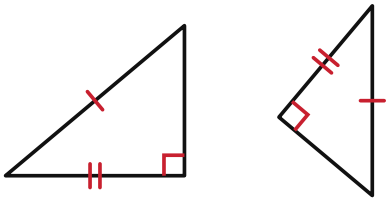
Criterion: _____

16.



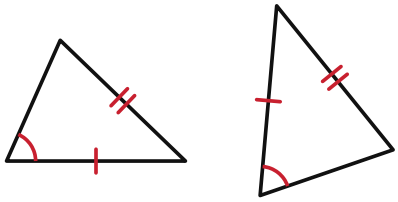
Criterion: _____

17.



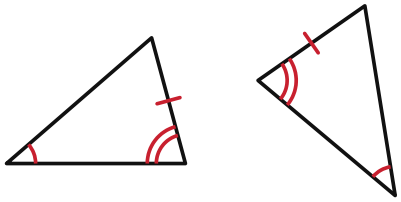
Criterion: _____

18.



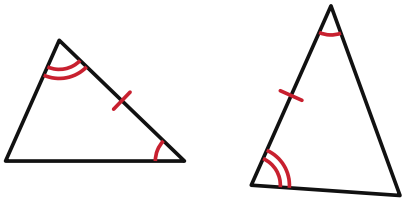
Criterion: _____

19.



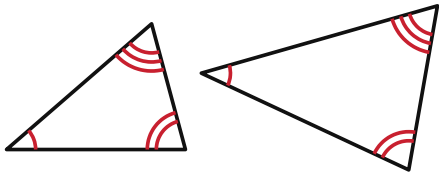
Criterion: _____

20.



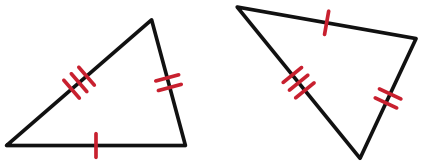
Criterion: _____

21.



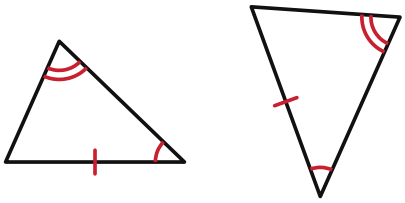
Criterion: _____

22.



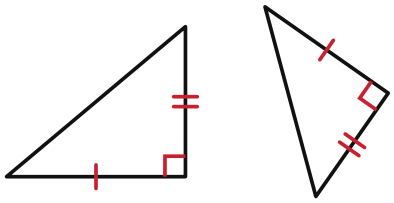
Criterion: _____

23.



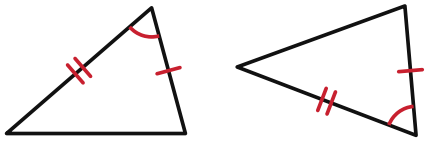
Criterion: _____

24.



Criterion: _____

25.



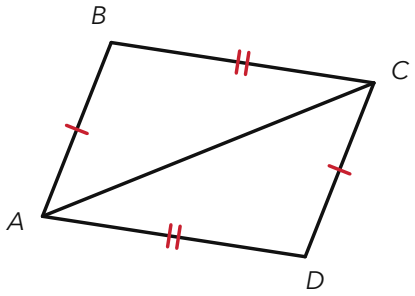
Criterion: _____

Two of these are marked SSA or AAA. Neither is a criterion – say so and move on.

Part 4. The part you have to notice

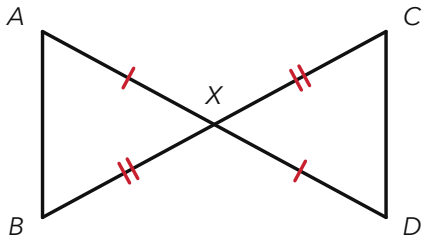
Not every congruent part is marked. A side in *both* triangles is congruent to itself, and vertical angles are always congruent. Find the unmarked part first, then name the criterion.

26.



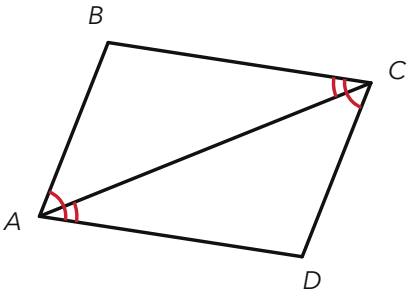
Criterion: _____

27.



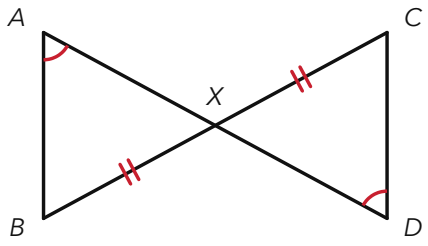
Criterion: _____

28.



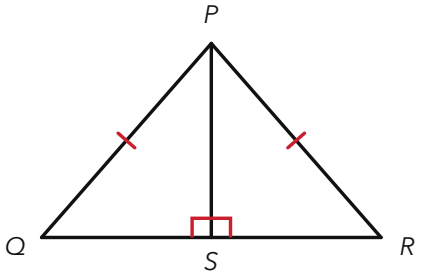
Criterion: _____

29.



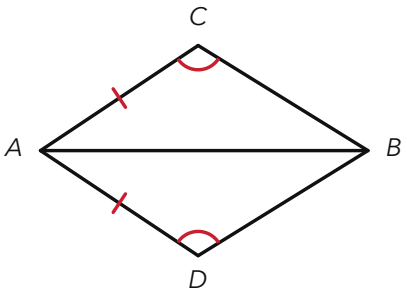
Criterion: _____

30.



Criterion: _____

31.

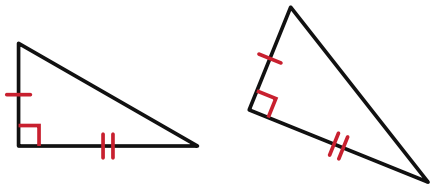


Criterion: _____

Part 5. Right triangles

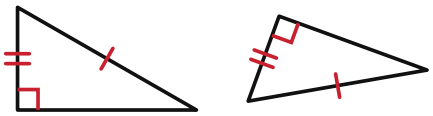
Every pair below is a pair of right triangles. HL is only one of the six answers – a right angle on its own proves nothing, and the hypotenuse is the side *opposite* it.

32.



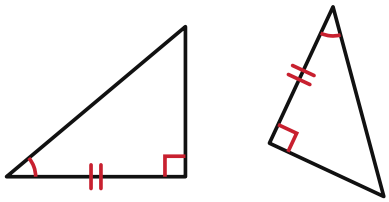
Criterion: _____

33.



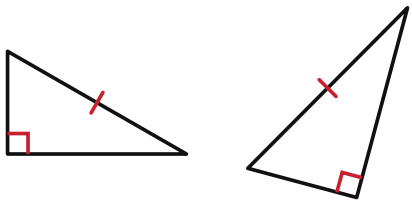
Criterion: _____

34.



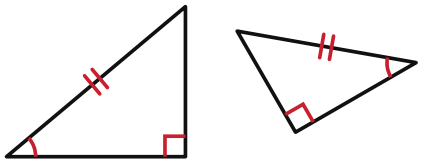
Criterion: _____

35.



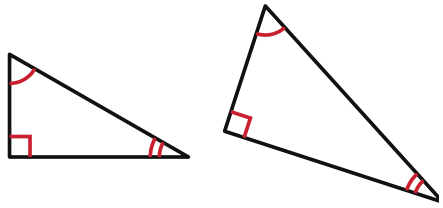
Criterion: _____

36.



Criterion: _____

37.



Criterion: _____

HL needs the right angle, the **hypotenuse**, and one leg. Two legs with the right angle between them is SAS – a true criterion, just not that one.